

Rigorous Notes on DPP, DPE and Optimal Stopping

1 Taxonomy of Controls in Stochastic Control Theory

We operate on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions, equipped with a d -dimensional standard Brownian motion $W = \{W_t\}_{t \geq 0}$. Let $U \subset \mathbb{R}^k$ be a separable metric space representing the control action space. In stochastic control, the type of control dictates the admissible information structure.

1.1 Open-Loop Controls

An open-loop control is a stochastic process $U : [0, T] \times \Omega \rightarrow U$ that is progressively measurable with respect to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$. Crucially, the control action U_t is adapted to the information available up to time t (the history of the driving noise) but does not depend explicitly on the concurrent state trajectory X_t .

1.2 Markovian (Closed-Loop / Feedback) Controls

A Markovian control policy is a deterministic, Borel measurable function $u : [0, T] \times \mathbb{R}^n \rightarrow U$. The control process is realized as $U_t = u(t, X_t)$. This structure restricts the controller's information to the current time and state. If u only depends on the state and not explicitly on time—that is, $u : \mathbb{R}^n \rightarrow U$ —the control is termed a **stationary Markovian control**.

2 Markov Properties of the Controlled SDE

Consider the controlled n -dimensional stochastic differential equation (SDE):

$$dX_t = b(t, X_t, U_t)dt + \sigma(t, X_t, U_t)dW_t \quad (1)$$

If U_t is a general open-loop control, X_t will generally not be a Markov process. To evaluate conditions under which X_t is Markovian, we assume U_t is a Markovian feedback control: $U_t = u(t, X_t)$. Substituting this yields the autonomous functions $\tilde{b}(t, x) = b(t, x, u(t, x))$ and $\tilde{\sigma}(t, x) = \sigma(t, x, u(t, x))$. The closed-loop SDE becomes:

$$dX_t = \tilde{b}(t, X_t)dt + \tilde{\sigma}(t, X_t)dW_t \quad (2)$$

2.1 Conditions for the Markov Property

For X_t to be a Markov process, the SDE must possess a unique weak solution (uniqueness in law) for every initial condition. This is guaranteed by the following measure-theoretic conditions:

1. **Joint Borel Measurability:** $\tilde{b}$ and $\tilde{\sigma}$ must be jointly measurable on $\mathcal{B}([0, \infty)) \otimes \mathcal{B}(\mathbb{R}^n)$ to ensure progressive measurability of the stochastic integrals.
2. **Local Boundedness:** For any finite $T > 0$ and compact $K \subset \mathbb{R}^n$, $\sup_{(t,x) \in [0,T] \times K} (|\tilde{b}(t,x)| + \|\tilde{\sigma}(t,x)\|) < \infty$.
3. **Weak Uniqueness (Stroock-Varadhan conditions):** Let $a(t,x) = \tilde{\sigma}(t,x)\tilde{\sigma}(t,x)^T$. If $(t,x) \mapsto a(t,x)$ is continuous, uniformly elliptic (i.e., $\sum_{i,j=1}^n a_{ij}(t,x)\lambda_i\lambda_j \geq c|\lambda|^2$ for $c > 0$), and $\tilde{b}(t,x)$ is bounded and measurable, the martingale problem for the generator A_t is well-posed. Uniqueness in the martingale problem ensures the finite-dimensional distributions are completely determined by the initial state, establishing the Markov property.

2.2 Conditions for the Strong Markov Property

A Markov process is Strong Markov if its memoryless property holds at any almost-surely finite stopping time τ . For continuous-path processes, this requires the Feller property, which typically relies on strong existence and uniqueness:

1. **Uniform Global Lipschitz Continuity:** There exists $K > 0$ such that $|\tilde{b}(t,x) - \tilde{b}(t,y)| + \|\tilde{\sigma}(t,x) - \tilde{\sigma}(t,y)\| \leq K|x - y|$. This guarantees pathwise uniqueness and continuous dependence on initial data.
2. **Linear Growth Bound:** There exists $C > 0$ such that $|\tilde{b}(t,x)|^2 + \|\tilde{\sigma}(t,x)\|^2 \leq C(1 + |x|^2)$. This prevents finite-time explosion, ensuring the transition semigroup is strictly conservative.

2.3 Conditions for Time-Homogeneity

A Strong Markov process is time-homogeneous if its transition kernel satisfies $P(s, x; s + h, B) = P(0, x; h, B)$. This requires:

1. **Stationary Policy:** The control must be a stationary Markovian feedback control, $U_t = u(X_t)$.
2. **Autonomous Plant Dynamics:** The drift and diffusion must have no explicit time dependence: $b(t, x, \alpha) \equiv b(x, \alpha)$ and $\sigma(t, x, \alpha) \equiv \sigma(x, \alpha)$.

When met, the SDE simplifies to $dX_t = \tilde{b}(X_t)dt + \tilde{\sigma}(X_t)dW_t$. Assuming the Lipschitz and linear growth conditions hold, X_t is a time-homogeneous Strong Markov process with a time-independent infinitesimal generator A .

3 The Dynamic Programming Principle

The primary goal of a stochastic control problem is to find an optimal control. In this section and the next, we introduce one of the main tools which allows us to achieve this goal: the dynamic programming equation (DPE). We first provide a heuristic derivation of the DPE associated with a stochastic control problem with infinite horizon discounted cost. As the motivation is merely formal, we make many assumptions which will need to be verified.

To simplify matters, we will work with the 1-dimensional controlled state process:

$$dX_t = b(X_t, U_t)dt + \sigma(X_t, U_t)dW_t, \quad X_0 = x \in \mathbb{R}, \quad (3)$$

where U is the control process with control space $U = \mathbb{R}$ and the state space is $S = \mathbb{R}$. We will restrict our attention to Markov controls. Recall that a control process U is a Markov control if it takes the form $U_t = u(X_t)$ for all $t \geq 0$ and some Markov control function $u : \mathbb{R} \rightarrow U = \mathbb{R}$. We assume that the state process $X \equiv X^{u,x}$ associated with Markov control function u is a time-homogeneous Markov process with generator $A \equiv A^u$:

$$Af(x) = b(x, u(x))f'(x) + \frac{1}{2}\sigma^2(x, u(x))f''(x),$$

for f in a suitable subset of $C^2(\mathbb{R})$. From the description of A , we see that the law of X is uniquely determined by the initial state x and the Markov control function u (and, of course, the given fixed functions b and σ).

We consider the infinite horizon discounted cost function:

$$J(x, u) = \mathbb{E} \int_0^\infty e^{-\gamma s} g(X_s, u(X_s)) ds.$$

(When the control process U is given by a Markov control function u , we often denote the associated cost by $J(x, u)$ rather than $J(x, U)$.) With b and σ fixed, the law of X , and hence the expected value above, is completely specified by x and u . To simplify notation we write $\mathbb{E}$ rather than $\mathbb{E}^{u,x}$.

The value function for this stochastic control problem is:

$$V(x) = \inf_u J(x, u),$$

where the infimum is taken over all Markov control functions. We assume that there exists an optimal Markov control function u^* which satisfies $V(x) = J(x, u^*)$ for all $x \in \mathbb{R}$. We denote by X^* the controlled process corresponding to u^* . Using the optimal Markov control function u^* , we have for $x \in \mathbb{R}$ and $t \in (0, \infty)$:

$$V(x) = J(x, u^*) = \mathbb{E} \int_0^t e^{-\gamma s} g(X_s^*, u^*(X_s^*)) ds + \mathbb{E} \int_t^\infty e^{-\gamma s} g(X_s^*, u^*(X_s^*)) ds.$$

Considering the second term on the RHS, we have:

$$\begin{aligned} \mathbb{E} \int_t^\infty e^{-\gamma s} g(X_s^*, u^*(X_s^*)) ds &= e^{-\gamma t} \mathbb{E} \int_t^\infty e^{-\gamma(s-t)} g(X_s^*, u^*(X_s^*)) ds \\ &= e^{-\gamma t} \mathbb{E} \int_0^\infty e^{-\gamma v} g(X_{v+t}^*, u^*(X_{v+t}^*)) dv \\ &= e^{-\gamma t} \mathbb{E} \int_0^\infty e^{-\gamma v} g(\tilde{X}_v^*, u(\tilde{X}_v^*)) dv. \end{aligned} \quad (4)$$

Above, we have used the change of variables $v = s - t$ and introduced the process $\{\tilde{X}_v^*, v \geq 0\}$, where $\tilde{X}_v^* = X_{v+t}^*$.

3.1 The Crucial Role of Time-Homogeneity

At this juncture, the assumption of *time-homogeneity* is the absolute linchpin that makes the dynamic programming argument mathematically viable for an infinite horizon.

First, time-homogeneity guarantees the existence of a stationary value function $V(x)$. Because the system parameters (plant dynamics, control policy, infinite horizon) are invariant under time translation, the optimal cost depends strictly on *where* the process is, not *when* it is there. If the system had explicit time dependence, we would be forced to define $V(t, x)$.

Second, and most critically in (4), time-homogeneity allows us to equate the laws of the shifted processes. Because X^* is a time-homogeneous Markov process, its transition kernel is invariant under time shifts: $P(t, X_t^*; t+v, \cdot) = P(0, X_0^*; v, \cdot)$. Thus, conditioned on the σ -algebra $\mathcal{F}_t$, the shifted process $\tilde{X}_v^*$ evolving from $\tilde{X}_0^* = X_t^*$ behaves probabilistically exactly like a brand-new process starting at time 0 from state X_t^* .

If $\tilde{X}_0^* = x$, then the processes X^* and $\tilde{X}^*$ have the identical law. In particular, given $\tilde{X}_0^* = x$, we have:

$$\mathbb{E} \left[\int_0^\infty e^{-\gamma v} g(\tilde{X}_v^*, u(\tilde{X}_v^*)) dv \mid \tilde{X}_0^* = x \right] = \mathbb{E} \int_0^\infty e^{-\gamma s} g(X_s^*, u(X_s^*)) ds = J(x, u^*),$$

which is equal to $V(x)$ by optimality of u^* . Using similar reasoning applied to the random starting state $\tilde{X}_0^* = X_t^*$, we collapse the tail integral into the value function:

$$\begin{aligned} \mathbb{E} \int_t^\infty e^{-\gamma s} g(X_s^*, u(X_s^*)) ds &= e^{-\gamma t} \mathbb{E} \left[\int_0^\infty e^{-\gamma v} g(\tilde{X}_v^*, u(\tilde{X}_v^*)) dv \right] \\ &= e^{-\gamma t} \mathbb{E} \left[\mathbb{E} \left[\int_0^\infty e^{-\gamma v} g(\tilde{X}_v^*, u(\tilde{X}_v^*)) dv \mid \tilde{X}_0^* \right] \right] \\ &= e^{-\gamma t} \mathbb{E}[J(\tilde{X}_0^*, u^*)] \\ &= e^{-\gamma t} \mathbb{E}[J(X_t^*, u^*)] \\ &= e^{-\gamma t} \mathbb{E}[V(X_t^*)]. \end{aligned}$$

Thus we have:

$$V(x) = \mathbb{E} \int_0^t e^{-\gamma s} g(X_s^*, u^*(X_s^*)) ds + e^{-\gamma t} \mathbb{E}[V(X_t^*)]. \quad (5)$$

Now let u be an arbitrary Markov control function. We construct a new Markov control function $\hat{u}$ which satisfies $\hat{u} = u$ from time 0 until time t and $\hat{u} = u^*$ afterwards. Letting X^u denote the associated controlled process, we have:

$$V(x) \leq J(x, \hat{u}) = \mathbb{E} \int_0^t e^{-\gamma s} g(X_s^u, u(X_s^u)) ds + \mathbb{E} \int_t^\infty e^{-\gamma s} g(X_s^u, u^*(X_s^u)) ds.$$

Using the same reasoning we used to derive (5), we obtain:

$$V(x) \leq \mathbb{E} \int_0^t e^{-\gamma s} g(X_s^u, u(X_s^u)) ds + e^{-\gamma t} \mathbb{E}[V(X_t^u)]. \quad (6)$$

Summarizing (5) and (6), we have arrived at the dynamic programming principle:

$$V(x) = \inf_u \mathbb{E} \left[\int_0^t e^{-\gamma s} g(X_s^u, u(X_s^u)) ds + e^{-\gamma t} V(X_t^u) \right], \quad (7)$$

where the infimum is over all Markov control functions.

4 Derivation of the Dynamic Programming Equation

Let u be an arbitrary Markov control with corresponding state process X and generator $A \equiv A^u$. Applying Itô's rule to the function $f(t, x) = e^{-\gamma t}V(x)$ we have:

$$e^{-\gamma t}V(X_t) = V(x) + \int_0^t e^{-\gamma s}[-\gamma V(X_s) + AV(X_s)]ds + \text{martingale},$$

where we have assumed that V is smooth enough and that the stochastic integral term is a martingale. Adding $\int_0^t e^{-\gamma s}g(X_s, u(X_s))ds$ and then taking expected values on both sides of the above equation we obtain:

$$\mathbb{E} \left[\int_0^t e^{-\gamma s}g(X_s, u(X_s))ds + e^{-\gamma t}V(X_t) \right] = V(x) + \mathbb{E} \int_0^t e^{-\gamma s}[-\gamma V(X_s) + AV(X_s) + g(X_s, u(X_s))]ds. \quad (8)$$

Now applying the dynamic programming principle (7) we see that the LHS of the above equation is no smaller than $V(x)$. Therefore:

$$0 \leq \mathbb{E} \int_0^t e^{-\gamma s}[-\gamma V(X_s) + AV(X_s) + g(X_s, u(X_s))]ds.$$

On the other hand, if u^* is an optimal Markov control function (which we assume exists), we have an equality:

$$0 = \mathbb{E} \int_0^t e^{-\gamma s}[-\gamma V(X_s^*) + A^*V(X_s^*) + g(X_s^*, u^*(X_s^*))]ds,$$

where X^* denotes the corresponding controlled process with generator A^* . Now divide by t and take limits as $t \downarrow 0$. Note that $X_t \rightarrow x$, and we assume that $u(X_t) \rightarrow u(x)$. Assuming that we can interchange the limit and the expected value and the various functions are smooth enough, we have:

$$0 \leq [-\gamma V(x) + AV(x) + g(x, u(x))]$$

for any Markov control function u , with equality in the case where the control is optimal:

$$0 = [-\gamma V(x) + A^*V(x) + g(x, u^*(x))].$$

We can summarize these last two displays by writing the equivalent statement:

$$0 = \inf_u [-\gamma V(x) + A^u V(x) + g(x, u(x))],$$

where, as usual, the infimum is taken over all Markov control functions. Now for fixed $x \in \mathbb{R}$, recalling the definition of A^u we see that the Markov control function u appears in the quantity to be minimized above only through the value $u(x) \in U$. Thus, we can write the above equation equivalently as:

$$0 = \inf_{\alpha \in U} [-\gamma V(x) + A^\alpha V(x) + g(x, \alpha)], \quad (9)$$

where

$$A^\alpha f(x) = b(x, \alpha)f'(x) + \frac{1}{2}\sigma^2(x, \alpha)f''(x).$$

Equation (9) is known as the dynamic programming equation (DPE).

5 Optimal Stopping Problems

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space equipped with filtration $\{\mathcal{F}_t\}$ and a d -dimensional $\{\mathcal{F}_t\}$ -Wiener process $\{W_t\}$. Starting from state $x \in \mathbb{R}^n$ at time $t \in [0, T)$, the state process X with state space $S = \mathbb{R}^n$ evolves according to the (uncontrolled) SDE:

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t, \quad X_0 = x. \quad (10)$$

As usual, b and σ are functions $b : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$ which satisfy the standard Lipschitz and linear growth conditions.

In the problem we consider here, we are not able to control the values of the state process directly. Rather, we are able to choose a time at which we stop accumulating running costs (and pay some terminal cost). The main requirement on our choice of time is that we cannot anticipate the future; the decision whether or not to stop should be based on current and past information alone. Mathematically speaking, we only consider stopping times with respect to our underlying filtration. Recall that a random variable τ taking values in $[0, \infty) \cup \{\infty\}$ is an $\{\mathcal{F}_t\}$ -stopping time if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$. We denote by $\mathcal{S}$ the set of all $\{\mathcal{F}_t\}$ -stopping times.

The cost associated with starting in initial state $x \in \mathbb{R}^n$ and stopping at time $\tau \in \mathcal{S}$ is:

$$J(x, \tau) = \mathbb{E} \left[\int_0^\tau e^{-\gamma s} g(X_s) ds + e^{-\gamma \tau} h(X_\tau) \mathbf{1}_{\{\tau < \infty\}} \right], \quad (11)$$

where g is the running cost function, h is the terminal cost function, and $\gamma > 0$ is the discount factor. The value function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ of the optimal stopping problem is:

$$V(x) = \inf_{\tau \in \mathcal{S}} J(x, \tau).$$

Suppose there exists a continuation region $D \subset \mathbb{R}^n$, an open set, such that $\tau_x^* = \inf\{t \geq 0 : X_t \notin D\}$ is optimal for all $x \in \mathbb{R}^n$, i.e., $V(x) = J(x, \tau_x^*)$.

Fix $x \in \mathbb{R}^n$ and let $\tau \in \mathcal{S}$ be arbitrary. Define $\tilde{\tau} = \inf\{t \geq \tau : X_t \notin D\}$. Following the stopping rule $\tilde{\tau}$, our decision is to continue until time τ , whether or not it is optimal to stop earlier; after time τ we then stop the first time the process exits D . Proceeding analogously to the infinite horizon case, since $V(x) \leq J(x, \tilde{\tau})$ we have:

$$\begin{aligned} V(x) \leq J(x, \tilde{\tau}) &= \mathbb{E} \left[\int_0^{\tilde{\tau}} e^{-\gamma s} g(X_s) ds + e^{-\gamma \tilde{\tau}} h(X_{\tilde{\tau}}) \mathbf{1}_{\{\tilde{\tau} < \infty\}} \right] \\ &= \mathbb{E} \left[\int_0^\tau e^{-\gamma s} g(X_s) ds + e^{-\gamma \tau} \int_\tau^{\tilde{\tau}} e^{-\gamma(s-\tau)} g(X_s) ds + e^{-\gamma(\tilde{\tau}-\tau)} h(X_{\tilde{\tau}}) \mathbf{1}_{\{\tilde{\tau} < \infty\}} \right] \\ &= \mathbb{E} \left[\int_0^\tau e^{-\gamma s} g(X_s) ds + e^{-\gamma \tau} J(X_\tau, \tau_{X_\tau}^*) \right] \end{aligned} \quad (12)$$

$$= \mathbb{E} \left[\int_0^\tau e^{-\gamma s} g(X_s) ds + e^{-\gamma \tau} V(X_\tau) \right], \quad (13)$$

where the last line follows from the optimality of the stopping rule $\tau^*(X_\tau)$: starting from state X_τ we are stopping at time $\tilde{\tau}$, the first time (after τ) that the state process exits the continuation region.

Of course, if the arbitrary τ satisfies $\tau \leq \tau_x^*$, then $\tilde{\tau} = \tau_x^*$, and so:

$$V(x) = J(x, \tau_x^*) = J(x, \tilde{\tau}) = \mathbb{E} \left[\int_0^\tau e^{-\gamma s} g(X_s) ds + e^{-\gamma \tau} V(X_\tau) \right], \quad (14)$$

where the last equality follows similarly to (13). Combining (13) and (14), we have the dynamic programming principle for optimal stopping:

$$V(x) = \inf_{\tau \in \mathcal{S}} \mathbb{E} \left[\int_0^\tau e^{-\gamma s} g(X_s) ds + e^{-\gamma \tau} V(X_\tau) \right]. \quad (15)$$

5.1 The Role of the Strong Markov Property

The transition from the shifted integral to the cost function evaluated at a random state in (12) relies heavily on the **Strong Markov property**. The expectation operator effectively conditions on the history of the process up to the random time τ (the σ -algebra $\mathcal{F}_\tau$). Because τ is an arbitrary stopping time and not a deterministic fixed time t , the standard Markov property is insufficient. To assert that the expected future cost from τ onward depends strictly on the state X_τ and is completely independent of the path taken to get there, the Strong Markov property must be invoked. It guarantees that the shifted process $X_{\tau+}$ behaves identically to a new Brownian motion starting at X_τ , permitting the substitution of the rigorous definition of J .

5.2 The HJB Equation for Optimal Stopping

Now we wish to obtain a dynamic programming equation. Again, fix $x \in \mathbb{R}^n$ and let $\tau \in \mathcal{S}$. Formally applying Itô's rule to $e^{-\gamma \tau} V(X_\tau)$ we obtain:

$$\mathbb{E}[e^{-\gamma \tau} V(X_\tau)] = V(x) + \mathbb{E} \int_0^\tau e^{-\gamma t} [AV(X_t) - \gamma V(X_t)] dt, \quad (16)$$

where as usual $AV(x) = b(x) \cdot \nabla V(x) + \frac{1}{2} \text{Tr}(\sigma(x)\sigma(x)^T \nabla^2 V(x))$ (when the various derivatives exist). First consider τ with $\tau \leq \tau_x^*$. Then (16) plus (14) yield:

$$0 = \mathbb{E} \int_0^\tau e^{-\gamma t} [-\gamma V(X_t) + AV(X_t) + g(X_t)] dt. \quad (17)$$

Suppose that $x \in D$. Since D is open and X has continuous sample paths, we have $\tau_x^* > 0$. Thus, dividing by τ and taking limits as $\tau \rightarrow 0$ in (17) yields:

$$0 = -\gamma V(x) + AV(x) + g(x), \quad x \in D. \quad (18)$$

If $x \notin D$, then $\tau_x^* = 0$, and so $\tau = 0$ and (17) does not yield any information. However, since τ_x^* is optimal, $V(x) = J(x, \tau_x^*)$, and $J(x, \tau_x^*) = h(x)$ by definition. So we have:

$$0 = h(x) - V(x), \quad x \notin D. \quad (19)$$

Now with $\tau \in \mathcal{S}$ arbitrary, (16) plus (13) yield (proceeding as we did to arrive at (18)):

$$0 \leq -\gamma V(x) + AV(x) + g(x), \quad x \in \mathbb{R}^n. \quad (20)$$

Also, since we can always stop immediately and pay the terminal cost (i.e., $V(x) \leq J(x, 0) = h(x)$ for any x):

$$0 \leq h(x) - V(x), \quad x \in \mathbb{R}^n. \quad (21)$$

Thus we have two inequalities (20), (21) which are satisfied for every x . In addition, we see from (18), (19) that for every x , at least one of these inequalities must hold with equality. We can summarize this as a variational inequality:

$$\min\{-\gamma V(x) + AV(x) + g(x), h(x) - V(x)\} = 0, \quad x \in \mathbb{R}^n. \quad (22)$$

The states in the continuation region are precisely those for which the equality is attained by the leftmost term:

$$D = \{x \in \mathbb{R}^n : h(x) - V(x) > 0\}.$$

5.3 Regularity of the Value Function and the Geometry of D

The derivation above relies on explicit assumptions regarding the regularity of the value function V and the terminal cost h , which strictly govern the geometry of the continuation region D .

First, for D to be mathematically defined as an open set, the function defining its boundary must possess specific topological properties. If the terminal cost h and the value function V are continuous, their difference $h(x) - V(x)$ is continuous. Because D is defined by the strict inequality $\{h - V > 0\}$, it is the preimage of the open interval $(0, \infty)$ under a continuous mapping. Therefore, the baseline requirement for D to be an open set is the continuity of V and h .

Second, there is a fundamental flaw in the naive assumption of classical C^2 regularity. The text formally applies Itô's rule to $V(X_t)$ in (16), which strictly requires $V \in C^2(\mathbb{R}^n)$. However, in optimal stopping problems, V is almost never globally C^2 . While V may be C^2 strictly inside the continuation region D and strictly inside the stopping region D^c , its second derivative $\nabla^2 V(x)$ typically exhibits a jump discontinuity exactly at the free boundary ∂D . Because of this lack of global C^2 smoothness across ∂D , the dynamic programming equation (22) must practically be understood in a weak (Sobolev) sense or the Viscosity sense, rather than the classical differential sense assumed by standard Itô calculus.